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Undergraduate Nursing Students' Perceptions, Needs, and Expectations Regarding Large Language Model–Based Virtual Patients: A Qualitative Study

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Abstract

Background: Communication competence is a core component of clinical competence among undergraduate nursing students. However, conventional communication teaching is often constrained by limited contextual authenticity, insufficient opportunities for repeated practice, and inadequate feedback support. Large language model-based virtual patients (LLM-VPs) offer a new approach to nursing communication training, but limited evidence exists on how undergraduate nursing students, based on their prior communication-learning experiences, anticipate the educational value, functional requirements, and boundaries of such systems.

Objective: This study aimed to explore undergraduate nursing students' anticipated perceptions, functional needs, preliminary acceptance, and views on potential risks and boundaries regarding the use of LLM-VPs for communication training before they had used a dedicated LLM-VP system. The findings were intended to provide early evidence for the integration of this technology into nursing education.

Methods: This study employed a descriptive qualitative design. A total of 15 undergraduate nursing students were recruited through convenience sampling. Data were collected through semi-structured interviews after written informed consent had been obtained from each participant. Data collection and analysis were conducted concurrently until theoretical saturation was reached. The collected data were coded and analyzed using thematic analysis, with the assistance of NVivo 14 software.

Results: Four main themes were identified: emerging educational gaps and competence needs; expected system form and design requirements; mechanisms underlying trust and sustained use; and perceived risks and boundaries of applicability. Overall, nursing students expressed cautiously optimistic expectations toward LLM-VPs and articulated diverse functional expectations for aligning these systems with nursing-specific learning needs.

Conclusions: From the anticipatory perspective of undergraduate nursing students, LLM-VPs have practical appeal and potential utility in nursing communication education. However, their educational value remains conditional and requires further

validation in real-use studies. Students expected LLM-VPs to function not as replacements for real clinical encounters but as supplementary training tools that support repeated practice, immediate feedback, and preclinical rehearsal. Future implementation should be grounded in nursing-specific adaptation, deep curricular integration, rigorous knowledge verification, ethical governance, and empirical evaluation of real-world educational effects to support standardized, cautious, and sustainable use in nursing education.

Keywords

Large language models; Nursing education; Student perceptives; Qualitative research

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Introduction

Effective patient communication is a foundational competency for ensuring patient safety, improving the quality of care, and enhancing patient outcomes[1, 2]. It should therefore be systematically strengthened during the training of health care professionals. According to Kolb's experiential learning cycle, learners can effectively acquire relevant skills only through concrete experience[3, 4]. In medical and nursing education, providing students with extensive opportunities to engage with authentic clinical situations is essential. Training involving patient encounters not only enhances nursing students' communication competence but also facilitates the development of their information-gathering skills[5]. At the same time, simulation-based education, as a cornerstone of medical and nursing training, provides learners with safe and repeatable opportunities for clinical practice by creating realistic practice environments, thereby strengthening practical skill development while reducing patients' exposure to potential risks[6]. Although simulation-based education has substantial pedagogical value, traditional simulation approaches still face multiple practical constraints. Conventional training modalities, such as interactions with real patients or standardized patients, often impose psychological pressure on learners and offer limited opportunities for repeated practice[7, 8]. In addition, these approaches are frequently constrained by several factors, including patient availability, the costs of recruiting and training actors, and the logistical demands of organizing such activities[9]. Other methods, such as peer role-play, are widely used because of their accessibility and feasibility; however, students often struggle to effectively simulate critical communication skills and emotional

expression because of limited knowledge, inadequate performance ability, and feelings of embarrassment[10].

In recent years, artificial intelligence (AI) has been reshaping medical and health professions education by advancing personalized learning through adaptive learning platforms, virtual simulation systems, and diagnostic tools, thereby optimizing instructional models, improving knowledge retention, and enhancing student learning outcomes[11, 12]. In particular, the development of large language models (LLMs) has enabled the generation of conversational responses that align closely with human cognitive processes through the modeling of grammatical and semantic patterns underlying human communication[8]. Moreover, in processing large volumes of medical literature and electronic health records (EHRs), LLMs have demonstrated the ability to extract detailed clinical information from patient discharge summaries[13]. In medical and nursing education, LLMs have broad potential applications, including self-directed learning, scenario-based simulation, and writing support. Functionally, they extend beyond the limitations of traditional learning approaches in 4 principal dimensions: temporal and spatial flexibility, learner engagement, efficiency, and personalized support[14]. Given these capabilities and advantages, LLMs may facilitate the implementation of high-fidelity simulation by enabling the development of virtual patients that can accurately emulate real patients in educational settings, thereby alleviating some of the challenges associated with traditional simulation and driving transformation in conventional teaching approaches.

At present, LLM-based virtual patients (LLM-VPs) has attracted substantial attention[15]. Existing studies have primarily focused on their feasibility in medical

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education and their potential to improve communication skills and learning outcomes. Overall, this body of research supports the value of LLM-VPs as an effective approach for communication training in medical students. In nursing education, similar positive effects on nursing students' nurse-patient communication have also been reported[5, 16, 17]. However, most prior research has focused on the effects of LLM-VPs on individual learning outcomes after implementation. Less attention has been given to the preimplementation perspectives of nursing students as intended end users, particularly their anticipated views, functional needs, and concerns shaped by prior communication-learning experiences before formal system development, optimization, and curricular integration.

For nursing students who have not yet used a dedicated LLM-VP system, their perspectives should not be interpreted as evaluations of actual system effectiveness or educational fit. Nevertheless, such perspectives can provide important early evidence for learner-centered design, context-sensitive development for nursing education, and subsequent implementation research. Without this preimplementation evidence, system functions may diverge from nursing students' actual learning difficulties, the core competencies targeted in nursing communication training, and the practical contexts in which such systems would be used. This mismatch may ultimately compromise future acceptability, educational relevance, and implementation sustainability.

The present qualitative study therefore explored undergraduate nursing students' perspectives on LLM-VPs. Because participants had not previously used LLM-VPs, the study did not aim to evaluate actual effectiveness. Instead, drawing on students'

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existing experiences with communication learning, it examined their anticipated perceptions, needs, expected value, preliminary acceptance, and potential concerns regarding the use of LLM-VPs for communication learning and training. Specifically, the study addressed four research questions: (1) What core difficulties and unmet needs do undergraduate nursing students experience in current communication learning and training? (2) Based on their prior learning experiences, what forms of support and key design features do undergraduate nursing students expect LLM-VPs to provide in future communication training? (3) What factors may influence undergraduate nursing students' preliminary trust, acceptance, and expectations for future use of LLM-VPs? (4) How do undergraduate nursing students anticipate the potential risks, ethical concerns, and boundaries of applicability of LLM-VPs in communication training?

By clarifying these preimplementation perspectives, this study may inform the learner-centered design, context-specific development, and preliminary curricular integration of LLM-VPs in nursing education. It also provides early empirical evidence for developing nursing communication training models that are grounded in learner needs and that better connect classroom learning with clinical practice.

Methods

Design

This study adopted a qualitative descriptive design[18] to explore undergraduate nursing students' perceptions of and expectations for the use of LLM-VPs in communication learning. Data were collected through face-to-face semistructured

interviews. This design was selected because it allows participants to provide relevant information in a structured manner while retaining the freedom to express their personal views[19], thereby generating credible data that closely reflect nursing students' authentic perspectives on the use of LLM-VPs for communication learning. The study was reported in accordance with the Consolidated Criteria for Reporting Qualitative Research (COREQ) checklist[20] (Supplementary Material 1).

Setting and Participants

Participants were recruited from Hangzhou Normal University in Hangzhou, China, between December 2025 and February 2026. In this qualitative study, purposive sampling was used to recruit participants who met criteria relevant to the study topic, thereby ensuring the relevance and depth of the sample. The final sample size was determined according to the principle of data saturation, that is, when no new information emerged and responses became repetitive[21].

The first author distributed recruitment information through online communication channels commonly used by students, including WeChat (Tencent) and DingTalk (Alibaba). The recruitment notice described the study purpose, eligibility criteria, interview arrangements, voluntary nature of participation, and researcher contact information. Students who were interested in the study contacted the first author directly. The first author then assessed eligibility and arranged the interview. Course instructors, college administrators, and teaching management staff were not involved in recruitment, eligibility screening, or interview scheduling and were not informed of participant identities. Before the interviews, all participants confirmed that they had not previously used LLM-VPs and had not participated in any formal learning,

training, or demonstration activities involving LLM-VPs. All recruited participants volunteered to take part in the study and were able to communicate in Mandarin Chinese. No eligible students who contacted the research team declined participation. The inclusion and exclusion criteria used for purposive sampling are presented in Textbox 1.

Inclusion criteria

- Undergraduate nursing students currently enrolled in a nursing program who had completed the theoretical component of the Fundamental Nursing course.
- Students with basic digital literacy or a general understanding of the internet.
- Individuals with adequate verbal communication skills who were able to clearly and authentically express their views.
- Willingness to participate in the study and provision of informed consent.

Exclusion criteria

- Individuals with mental disorders that might impair their ability to respond to interview questions appropriately.
- Participants who withdrew from the research for any reason.

Textbox 1. Eligibility criteria for participants.

Data Collection

Data were collected through semistructured interviews conducted face-to-face by 1 researcher who had received formal training in qualitative research and had prior interviewing experience. We adopted the interview framework proposed by Krueger

and Casey[22], and, in line with the study objectives, developed a preliminary interview guide based on a review of the literature, discussions within the research team, and consultation with qualitative research experts. The guide was then pilot-tested with 3 undergraduate nursing students; data from the pilot interviews were not included in the formal study. The interview questions were subsequently refined on the basis of participant feedback and the pilot interview content, and the final interview guide was established (Supplementary Material 2).

All interviews were conducted in a quiet and private environment to ensure continuity of thought and minimize interruptions. Before each interview, the researcher reiterated the study purpose and interview content and obtained both verbal and written informed consent, including permission for audio recording. Before the formal interview began, the researcher first asked whether participants had heard of or understood LLM-VPs. The researcher then provided all participants with a standardized conceptual explanation covering the basic meaning of LLM-VPs, their main functions, and their potential application scenarios in nursing communication training. A brief comprehension check was conducted before the interview proceeded. During the interviews, the researcher used responsive, interpretive, summarizing, and reiterative techniques to encourage participants to share their views and expectations regarding the use of LLM-VPs in communication learning. All interviews were audio-recorded anonymously.

Data Analysis

Data were analyzed using inductive thematic analysis[23]. The analysis team comprised nursing professionals with diverse research expertise, including nursing

informatics, nursing education, and psychological and mental health nursing, and this supported a broader interpretation of the data[22, 24]. First, the first author transcribed the original audio data verbatim using a speech-to-text tool. The transcripts were then checked independently by 2 authors for accuracy and compiled into Word documents. These documents were subsequently imported into NVivo 14 (Lumivero) for further analysis.

The authors repeatedly read the transcripts in detail to obtain an overall understanding of the interview data. The data were then examined carefully, with individual concepts or sentences selected as meaning units. After the original expressions had been identified, the authors condensed and coded the data, grouped codes with similar meanings, and assigned descriptive labels. Any disagreements were resolved through discussion between the 2 authors, with reference to the original audio recordings when necessary. The entire coding process was supervised by another researcher to enhance credibility, and this researcher also participated in random checks of the analysis to ensure consistency.

The resulting categories were then further merged into higher-order categories, with continuous attention to their alignment with the research questions. Each research question was analyzed separately. When new insights emerged, they were discussed in team meetings, and revisions were proposed until consensus was reached among all authors.

Rigor and Reflexivity

To ensure methodological rigor, we followed the 4 criteria commonly used in qualitative research: credibility, dependability, confirmability, and transferability[25]. Details are presented in Table 1. Reflexivity was maintained throughout data collection, analysis, and reporting to continuously examine the potential influence of the researchers' disciplinary backgrounds, prior assumptions, and research positions on data generation and interpretation. Reflective journals and analytic memos were used to document the evolution of research decisions and analytic thinking throughout the study. All interview data were read repeatedly, and interpretations were continually examined, revised, and refined through ongoing discussions within the research team during coding and category development.

Using the memo and log functions in NVivo, the researchers documented the research process in detail, including reflections, ideas, personal beliefs, and judgments, to identify any unconscious biases that might have influenced the findings. The findings were reviewed repeatedly to ensure that they were not unduly shaped by individual characteristics or subjective judgments. In addition, during team discussions, members of the research team were able to question and challenge one another's interpretations, thereby facilitating collaborative reflexivity[26].

Quality criteria	Action taken by the researchers
Credibility	• Undergraduate nursing students were purposively selected as the target population. • Semistructured interviews, together with probing and

	clarification, were used to generate rich, in-depth data.
	• Regular discussions with coauthors were conducted to review codes and emerging categories.
Dependability	• The study design, sampling strategy, data collection, and analytic procedures were described in detail. • An interview guide was used to ensure consistency while allowing flexibility during interviews. • Reflective journals and analytic memos documented research decisions and analytic development, providing an audit trail.
Confirmability	• Reflexivity was maintained throughout the research process to examine the potential influence of researcher positions and assumptions on data generation and interpretation. • Coding and category development were repeatedly discussed within the research team to assess the credibility of alternative interpretations.
Transferability	• The research context, participant characteristics, and study procedures were described in sufficient detail. • Interview data were transcribed verbatim to enhance data accuracy.

Table 1. Methodological and analytical rigour of the qualitative data as recommended by Lincoln and Guba.

Ethics

The study was carried out in accordance with the ethical standards established in the Declaration of Helsinki. This study was approved by the Ethics Committee of School of Public Health and Nursing, Hangzhou Normal University (No. 2026009). Before the interviews, the researcher explained the study purpose, interview content, use of audio recordings, data storage procedures, and participant rights, and obtained written informed consent. All data were deidentified by replacing names with study codes, and identifiable information was removed or masked. Audio recordings and transcripts were stored in encrypted files accessible only to the research team. Findings were reported in aggregate form. Participation was entirely voluntary. Students could refuse participation or withdraw from the study at any time without providing a reason. Participation, nonparticipation, or withdrawal had no effect on course grades, academic evaluation, practicum or internship opportunities, or relationships with instructors, the college, or the university. Course instructors and teaching management staff were not involved in recruitment or interviews and were not informed of participant identities.

Results

Participant Characteristics

A total of 15 undergraduate nursing students participated in the study, and all interview data were included in the analysis. The total interview time was 431 minutes, with individual interviews ranging from 22 to 36 minutes and a mean duration of 29 (SD, 4.3) minutes per participant. Participants' demographic characteristics are presented in Table 2. Of the 15 participants, 5 were male and 10

were female. Participants ranged in age from 18 to 24 years, with a mean age of 21.3 (SD, 1.3) years. 4 participants were in the second year, 4 were in the third year, and 67 were in the fourth year of study. Eleven participants had observational placement experience, and 7 had clinical internship experience; among them, 7 had both observational placement and clinical internship experience.

Characteristics	Participants (N=15)
Age (years), mean (SD)	20.5 (5.0)
Sex, n(%)	
Male	5 (33)
Female	10 (67)
Academic Year, n (%)	
Second-year	4 (27)
Third-year	4 (27)
Fourth-year	7 (46)
Clinical Exposure	
Observational placement	
Yes	11 (73)
No	4 (27)
Clinical internship	
Yes	7 (47)
No	8 (53)

Table 2. Participants characteristics.

Thematic Overview of the Interview Data

Analysis of the interview data yielded 4 major themes: (1) emergence of pedagogical gaps and competency needs (Drivers), (2) expectations for the ideal system form and design requirements (Needs), (3) mechanisms shaping trust and sustained use (Willingness), and (4) perceived risks and boundaries of applicability (Challenges).

Overall, undergraduate nursing students' views of LLM-VPs showed a clear dual character. On the one hand, participants explicitly recognized substantial gaps in current communication learning and training and regarded LLM-VPs as a promising digital training tool with the potential to bridge the divide between classroom learning and clinical practice. On the other hand, they did not express unconditional optimism toward the technology. Instead, they simultaneously highlighted important limitations related to realism, knowledge accuracy, ethical governance, technological maturity, and the educational boundaries of appropriate use. The thematic overview diagram is shown in

Figure 1, and an overview of the coding framework is presented in Table 3.

Illustrative participant quotations organized by themes and subthemes are provided in Supplementary Material 3.

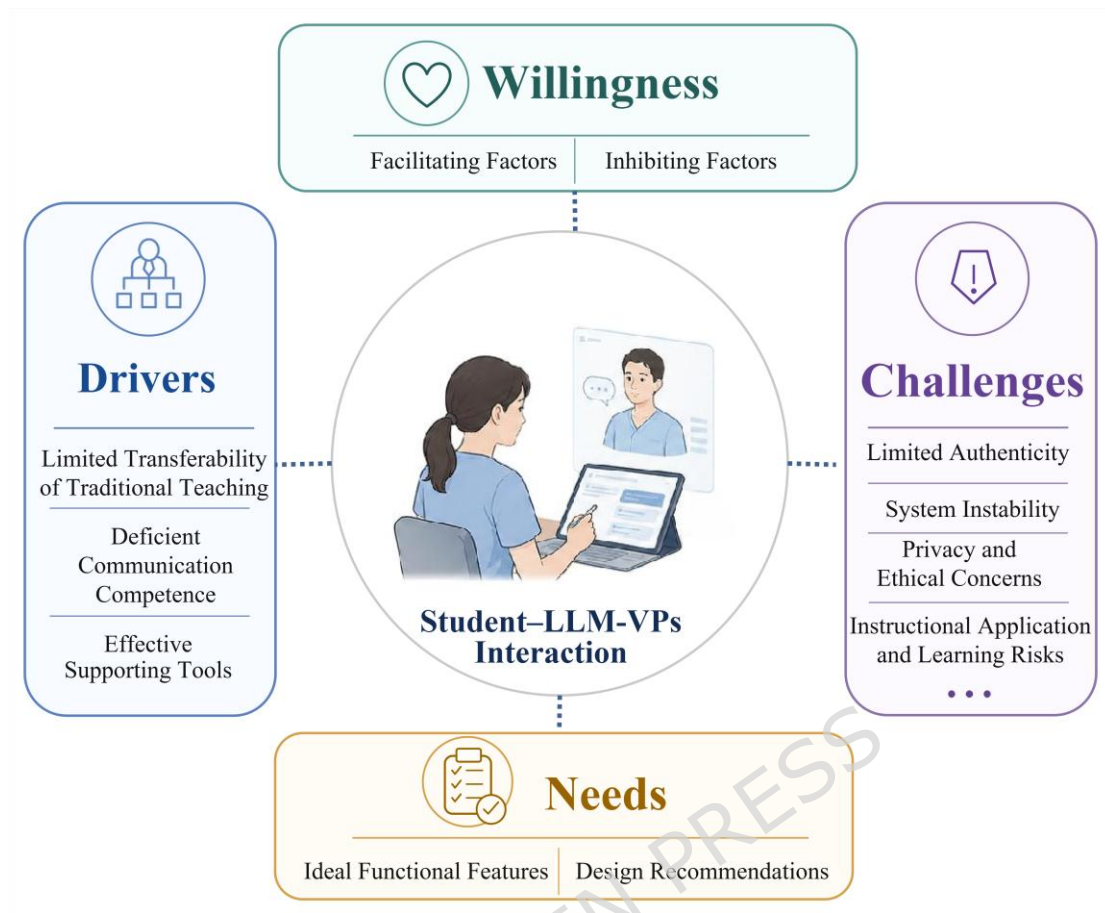


Figure 1. Overview Diagram of Thematic Information.

Theme	Sub-themes	Codes
Theme 1. Emergence of pedagogical gaps and competency needs	Limited transferability of traditional teaching to real clinical practice	Insufficient realism in simulation-based teaching; procedural and rigid communication training; inadequate exposure to complex clinical situations; limited opportunities for hands-on and immersive practice; insufficient feedback and constrained instructional effectiveness
	Contextual gaps in nursing students' clinical	Disorganized communication structure; difficulty initiating expression; insufficient emotional responsiveness; limited

	communication competence	situational adaptability; inadequate practical support
Theme 2.	Realistic and multifaceted	Diverse clinical roles and scenario simulation; multimodal
Expectations for	interactive experiences	and humanlike communication experiences; representation
the ideal system		of emotion and nonverbal communication; high realism and
form and design		immersive experiences
requirements		
	Intelligent learning and	Immediate feedback and debriefing support; adaptive
	assessment mechanisms	difficulty and stratified training; modular training and
		process guidance; knowledge support and learning record
		management
	An educational product	System stability; ease of operation; affordability;
	feasible for real-world	curriculum-embedded integration; accurate knowledge
	implementation	content
Theme 3.	Perceived benefits as a	Diversified training opportunities; improvement in
Mechanisms	driver of use	communication competence; pre-internship preparation;
shaping trust and		support for examination review
sustained use		
	Threshold factors	Hesitation caused by insufficient realism; system cost;
	constraining use	technical and accessibility barriers; time and effort burden
	Dual effects of external	Teacher encouragement; curricular alignment; passive use

	drivers and mandatory use mechanisms	driven by assessment requirements; institutional rules shaping adoption; tension between autonomy and compulsion; inhibition caused by negative evaluations
Theme 4. Real-world boundaries of implementation	Inherent limits in simulating real patients	Mechanical and procedural interaction; insufficient representation of complex emotions and subjectivity; difficulty reproducing patients' nonprofessional expressions; limited support for cultivating empathy and body-language awareness
	Knowledge accuracy as a determinant of educational value	Insufficient model performance; outdated knowledge; inadequate verification; risk of fabrication
	Implementation constrained by both ethical and technical conditions	Concerns about data privacy; requirements for ethical governance; risks related to technical instability
	Effects on learning behaviors in educational use	Dependence on virtual environments; erroneous expectations caused by AI accommodation; superficial implementation due to lack of teacher-led curricular integration

Table 3. Overview of coding framework.

Emergence of Pedagogical Gaps and Competency Needs

This theme captures nursing students' reflections on the limitations of traditional communication teaching and on their own communication difficulties. Participants suggested that although current approaches provide some basic training value, they are insufficient to meet the demands of authentic clinical communication development. They also described their communication challenges not as a single problem of "poor communication skills," but as a layered combination of cognitive, emotional, and practical difficulties.

1. Limited Transferability of Traditional Teaching to Real Clinical Practice

Participants reported that communication teaching in nursing programs was mainly delivered through classroom role-play, with teachers or students acting as patients, the use of standardized patients, or in some cases lecture-based teaching supported only by PowerPoint slides. They perceived these approaches as limited by insufficient clinical realism, restricted opportunities for repeated practice, and inadequate feedback and instructional guidance. As a result, the knowledge and skills acquired in training were not easily transferred to authentic clinical settings.

"Most of the simulation exercises carried out in class involve students acting with each other, which makes it very easy to lose the sense of realism." (N9)

"In class, there are only one or two cases before it ends, so opportunities for actual practice are very limited." (N15)

“...After the simulation, judging from the teacher’s feedback, they simply did not know what we were talking about, nor could they understand what we were doing, and we also did not know why they could not understand what we were doing... In the end, we just did not know what problems we had.” (N11)

2.Contextual Gaps in Clinical Communication Competence

Participants’ self-perceived communication difficulties extended beyond verbal expression alone and spanned 3 interrelated dimensions: logical organization, emotional responsiveness, and management of complex situations. Specifically, they described difficulties in asking structured questions, managing psychological tension during real interactions, and responding appropriately in emotionally charged, complex, or professionally demanding situations.

“My questioning is not very logical—it tends to be rather disorganized, and I may miss some very important questions. ... When faced with some of the patient’s emotions, I may not know how to handle them.” (N5)

“When I encounter patients with a less-than-friendly attitude, I hope I can respond more smoothly and confidently.” (N8)

Expectations for the Ideal System Form and Design Requirements

Participants’ expectations for LLM-VPs extended beyond a simple “conversational virtual patient.” Instead, they envisioned a comprehensive learning system that could integrate realistic scenario simulation, personalized feedback, knowledge support, and

opportunities for sustained practice. In their view, an ideal LLM-VP should resemble real clinical encounters while also functioning as an effective educational tool; it should deliver accurate content while remaining easy to use; and it should not only enable practice but also identify and correct mistakes.

1. Realistic and Multifaceted Interactive Experiences

Participants were generally dissatisfied with static, single-format, and highly templated patient simulations. They expected the system to provide training targets and interaction processes that more closely reflected clinical complexity. Realism, in their view, should include diversity in specialties, diseases, personalities, age groups, and stages of illness, as well as family communication scenarios and differences in expression associated with educational background, including dialect variation.

Participants also emphasized that patient responses should not appear overly idealized or unrealistically cooperative; instead, language should more closely resemble the direct, vague, or partially uncooperative communication often encountered in real hospital settings.

“There should be multiple types of patients—for example, different diseases, ages, and personalities should be available to choose from.” (N10)

“It should be more like a real patient. Its answers or understanding of questions should not be too perfect, because this is actually the case in most real clinical settings.” (N6)

In addition, participants preferred multimodal interaction rather than a single communication format. They believed that features such as voice-based communication, humanlike avatars, facial expressions, microexpressions, and emotionally responsive behaviors would substantially enhance immersion and realism. Notably, preferences regarding appearance were not uniform. Although most participants favored humanlike representations, some considered cartoon-style avatars easier to accept and less anxiety-provoking. This suggests that a switchable appearance style may be a more appropriate design option.

“It would be best if it could be a bit more emotional and expressive. The simulated patient could also have micro-expressions, so that a sense of immersion can be achieved.” (N12)

“If the patient is presented as a cartoon character, it might make me more willing to open and use the system, and it would help me feel more relaxed.” (N6)

2. Intelligent Learning and Assessment Mechanisms

Participants emphasized that the system should not be limited to dialogue practice alone, as such functionality would be too narrow. First, they expected the system to support modular learning across domains such as internal medicine, surgery, obstetrics, and pediatrics. In terms of training pathways, they expected progression from simple to complex tasks, with training stratified according to learner level or areas of weakness, thereby enabling adaptive difficulty adjustment. They also expected the system to save training records and display trajectories of progress,

thereby supporting a more intelligent form of technology-enabled learning. Second, participants expected personalized feedback, including identification of problems, summarization of omissions, and generation of performance reports. They also expected the system to be aligned with course materials and linked to relevant clinical guidelines, thereby creating a closed learning loop of practice, feedback, remediation, and repeated practice.

“I hope there could be different modules, such as internal medicine and surgery, so that everyone can choose according to their own needs.” (N10)

“It should start from the simple level first, and then, after I have adjusted my thoroughness of thinking and the logic of my speaking, the difficulty can gradually be increased.” (N7)

“It should give me some kind of feedback or summary. If it does not give me a summary, then I do not know where the problems are.” (N11)

At the same time, some participants noted that real clinical environments do not adapt to learners. This suggests that although students recognized the educational value of stratified training, they were also concerned that excessive adaptivity might weaken the unpredictability that characterizes authentic clinical communication.

“Not really, because clinical patients will not change because of you, and there is a great deal of uncertainty.” (N4)

3. An Educational Product That Is Feasible for Real-World Implementation

Participants were also concerned with whether the system would be genuinely trustworthy, usable, and sustainable. All content provided to students, in their view, needed to be professionally accurate to avoid misleading learning. At the same time, the actual user experience needed to be of high quality, including a clean interface, smooth operation, technical stability, and an affordable or preferably free pricing model. Participants also expected the system to align with course content and teaching materials. In their view, only a system that combined educational effectiveness, content reliability, and practical accessibility could realistically become part of routine learning.

“First of all, it has to ensure the accuracy of the knowledge points. Otherwise, if what we learn is all wrong, then it is meaningless... It would be best if it corresponded to our textbook content.” (N10)

“First of all, it should ideally be free and available for unlimited use. The system must run smoothly without lagging. ... It would be best if system malfunctions did not occur.” (N15)

Mechanisms Shaping Trust and Sustained Use

Overall, participants expressed a certain degree of willingness to use LLM-VPs, but this willingness was clearly conditional rather than unconditional. Adoption was shaped jointly by perceived learning benefits, integration into courses and

examinations, financial and time costs, system realism, and the overall user experience.

1. Perceived Benefits as a Driver of Use

Participants were more willing to use the system when it was perceived as more engaging than traditional teaching and when it appeared helpful for improving communication, preparing for clinical practice, reviewing for examinations, or completing course tasks. Willingness to use the system also increased when participants perceived that it could provide diverse training content. These findings suggest that use intentions were driven by both competency-development goals and performance-oriented goals.

“I would be quite willing to use it before going into clinical observation or internship. If it could also match the content taught in class, then there would be no need to listen to those boring PPTs or to do simulated dialogues with classmates.” (N1)

“I would also be willing to use it during the final exam period, because I could review and familiarize myself with things in advance, which would make me feel more confident about the exam.” (N12)

“It can provide many different scenarios for practice, so it would be quite helpful, and that would make me want to use it.” (N3)

2.Threshold Factors That Constrain Use

Actual adoption and sustained use were seen as strongly dependent on practical barriers. Economic cost was a central concern, with students emphasizing that, as an educational tool, the system should ideally be free of charge. Ease of use and time efficiency were also important; participants preferred a system that was easy to learn and required minimal effort to operate. Finally, the realism of the simulation was regarded as a key determinant of whether long-term use would be worthwhile. Thus, adoption of LLM-VPs was not determined simply by whether such a system existed, but by whether it was genuinely both useful and easy to use.

“If it requires payment, I would have some reservations about it. Also, if it is too complicated to operate, I would be quite resistant to using it. Another point is that if what it simulates differs greatly from real patients, then it would not be very meaningful.” (N8)

3.The Dual Effects of External Drivers and Mandatory Use Mechanisms

Participants’ intentions were also influenced by external drivers such as course assignments, examination pressure, and teacher requirements. Some stated that they would use the system if it were incorporated into course tasks, assignments, grading, or group presentations, whereas their proactive willingness would decline in the absence of such institutional arrangements. Teacher recommendations and peer evaluations were also perceived as factors that could either encourage or discourage use. Importantly, mandatory mechanisms were not viewed as uniformly beneficial. Participants indicated that if the system remained immature or offered a poor user

experience, but was implemented in an overly rigid manner, it could provoke passive compliance or even resistance.

“I probably would not use it on my own initiative. I would only use it if it were required to exam grades, or if the teacher specifically required us to use it.” (N11)

“If this were part of a course learning task assigned by the teacher, then I would be willing to use it. But after class, my willingness to use it might not be very high.” (N15)

“You would still have to look at everyone’s evaluations. If both classmates and teachers gave it very good feedback, then I might also be willing to give it a try.” (N10)

Real-World Boundaries of Implementation

Overall, although participants recognized the potential value of LLM-VPs for communication training, they were also acutely aware of substantial challenges related to realism, knowledge accuracy, technical stability, data ethics, and educational fit. In other words, LLM-VPs were not viewed as inherently effective teaching tools; rather, their educational value was considered highly dependent on the quality of system design, mechanisms for content governance, and the extent to which they were meaningfully integrated into curricula.

1. Inherent Limits in Simulating Real Patients

Most participants pointed out that the system might have clear limitations in reproducing real patients with high fidelity. In their view, because LLM-VPs are ultimately AI-based systems, they are unlikely to fully replicate the subjectivity, emotional richness, and highly variable interactive responses of real patients. Participants also worried that the system might adopt a mechanical, stereotyped, and overly procedural question-and-answer style, lacking the everyday language, narrative texture, and interpersonal warmth found in real communication, thereby weakening immersion and authenticity. This concern was especially salient in relation to humanistic care. Participants emphasized that empathy is not conveyed solely through verbal responses, but also through bodily movement, facial expression, and the immediacy of in-person presence—precisely the elements that virtual systems may struggle to represent adequately. These views suggest that nursing students regarded LLM-VPs more as a supplementary simulation tool for nurse-patient communication than as a full substitute for real patient encounters.

“It is mechanical and rigid, and the simulation is not realistic. ... It is also somewhat difficult to cultivate empathy, because this is a virtual patient. Empathy is easier to develop only through communicating and interacting with real people who have genuine emotions.” (N5)

2. Knowledge Accuracy as a Determinant of Educational Value

In educational settings, participants were highly sensitive to the accuracy and timeliness of system content. They worried that insufficient model performance,

system defects, or delayed updating of the knowledge base relative to current standards could mislead learning. Under such circumstances, the system would not only compromise learning effectiveness but would also rapidly erode students' trust.

“If a lower-performing model is used, because of the lag in its knowledge, some nursing knowledge may not keep up with the latest standards, and this may instead hinder learning.” (N14)

“Personally, I might be quite worried about the accuracy of the information. After all, it is an AI-type product, so it may fabricate things that do not actually exist.” (N9)

3.Implementation Constrained by Both Ethical and Technical Conditions

Participants believed that practical implementation of LLM-VPs would be constrained by multiple real-world conditions. First, if system training relied on real clinical cases, issues related to case access, patient authorization, ethical review, and privacy de-identification would be unavoidable. Second, the technical stability and maturity of the system itself were seen as critical determinants of practical usability. Thus, for LLM-VPs to be widely adopted in teaching, they would need not only to address privacy and data security risks but also to satisfy the basic requirements of educational settings for stability, speed, reliability, and sustainable operation.

“From a technical perspective, the corpus used to pretrain a large language model would definitely need to be based on real cases. These real cases would have to be obtained from hospitals. Ethical approval would certainly be required, and the data

would need to be de-identified, but even then, privacy leakage still cannot be completely ruled out.” (N8)

“The system may not be smooth enough and may lag during use. If it is not mature enough, I think it would be very difficult for it to be widely adopted.” (N1)

3. Effects on Learning Behaviors in Educational Use

Beyond technical concerns, participants also identified potential negative effects on student learning behaviors. Some suggested that if the system were overly friendly or excessively accommodating, students might develop unwarranted confidence or make incorrect judgments. Others worried that prolonged exposure to a relatively low-risk and controllable virtual environment might gradually reduce students' willingness to engage with real patients or lead them to adopt communication styles that were overly influenced by AI-generated language. Participants also emphasized that without careful integration by teachers in terms of curriculum design, training objectives, and assessment methods, even a technically functional system might result in merely superficial or symbolic implementation. This suggests that the challenges perceived by nursing students extend beyond whether the technology can be built to whether it can be used appropriately in educational practice.

“AI is relatively idealized, and the patients it simulates are also comparatively easier patients. Then, when students actually encounter real patients, they may develop a sense of resistance.” (N1)

“Integrating this system into nursing education also requires teachers to have a strong grasp of curriculum design and a clear planning framework. Otherwise, it will only remain superficial and become a mere formality.” (N14)

Discussion

This qualitative study explored undergraduate nursing students’ anticipated perceptions of using LLM-VPs for communication learning and training. Participants’ views did not reflect simple acceptance or rejection. Rather, they indicated a form of conditional acceptance shaped by unmet needs in current communication training, the perceived credibility of the system, and the extent to which LLM-VPs could be meaningfully integrated into nursing curricula. Because participants had not used a dedicated LLM-VP system, the value they described should be interpreted as perceived potential value rather than demonstrated educational effectiveness. Students generally viewed LLM-VPs as a potentially accessible, repeatable, low-risk, and feedback-oriented transitional learning environment that could help bridge classroom instruction and clinical practice, rather than as a substitute for teachers, standardized patients, or real patient encounters.

The demand for LLM-VPs expressed by nursing students can be understood as a response to structural limitations in conventional communication training. Difficulties in organizing clinical information, responding to patients’ emotions, and managing complex interpersonal situations may be related to limited opportunities for authentic, frequent, low-risk, and repeatable practice, as well as insufficient formative feedback. Nursing education has long faced a substantial gap between the theoretical knowledge students acquire in academic settings and the practical skills required in authentic

clinical environments, partly because of reduced learning opportunities and the lack of structured frameworks for translating knowledge into practice[27, 28]. According to Kolb's experiential learning theory[3, 4], the development of communication competence in nursing depends on repeated cycles of concrete experience and active experimentation. Traditional communication teaching, however, often lacks the lived experience of illness and the emotional depth of authentic encounters, leaving learners disengaged and limiting deeper reflection. In addition, differences in background and developmental goals may create discrepancies between teachers' and students' perceptions of instructional needs, which may in turn affect educational outcomes[29]. Taken together, the problems reported by students should not be understood merely as deficits in individual expressive ability, but rather as learner-level manifestations of the limitations of traditional teaching.

Course structure, teaching methods, and simulation fidelity jointly shape students' self-perceived readiness for practice and their ability to transfer learning into practice[26]. Accordingly, students expected LLM-VPs to function not merely as conversational agents but as educational systems capable of supporting experiential, reflective, and progressive learning. First, their potential effectiveness in nursing education depends heavily on nursing-specific relevance and contextual authenticity. Nursing communication involves not only information gathering but also emotional reassurance, health education, family communication, facilitation of adherence, and humanistic care[30, 31]. Students consequently expected LLM-VPs to support diverse patient scenarios and to use multimodal features to approximate clinical language, patient responses, and nonverbal cues. They also considered LLM-VPs attractive because such systems could integrate several elements that are often fragmented in

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traditional communication training: scenario exposure, repeated practice, immediate feedback, and self-correction. In this sense, an ideal LLM-VP system would support a continuous learning loop of practice, feedback, remediation, and further practice. Such a design may help address students' perceived lack of preparedness for clinical work, which is often intensified by fragmented curricula, inconsistent teaching strategies, and insufficient integration between academic learning and clinical practice[32].

The findings also suggest that nursing students' anticipated adoption of LLM-VPs is strongly pragmatic and task oriented. Their willingness to continue using such systems does not arise automatically, but is built on explicit evaluations of expected benefits and practical conditions. Their willingness to use such systems was not automatic; it depended on perceived benefits and realistic conditions of use. This pattern can be interpreted through concepts from the Unified Theory of Acceptance and Use of Technology (UTAUT)[33]. When students perceive performance expectancy—that is, when they believe the system can provide tangible, transferable, and educationally meaningful benefits—positive use intentions are more likely to emerge. Effort expectancy, reflected here in concerns about cost, convenience, and the burden of learning to use the system, indicates that nursing students already operate under substantial academic load[34, 35]. Under such circumstances, any educational technology that demands additional cognitive effort or involves complicated operating procedures is likely to encounter resistance in practice. At the same time, curriculum embedding and teacher or peer evaluation—forms of social influence—appeared to have dual effects. These factors may function as facilitating conditions, but they may also become sources of passive compliance if imposed

rigidly. This distinction is important. When an interactive system is combined with appropriate pedagogical strategies, it can enhance educational realism and promote reflection[36]. Moreover, when incorporated into a well-designed curriculum, high-fidelity simulation may provide greater educational benefit to learners[37, 38].

Therefore, the effective application of LLM-VPs in nursing education depends not only on the system itself but also on institutional arrangements and flexible instructional design by teachers. More importantly, in simulation-based educational technologies such as LLM-VPs, system realism and content accuracy constitute the foundation of trust. Students are more likely to engage with these systems when they perceive the scenarios as clinically realistic and the responses as professionally accurate. Sustained adoption is therefore unlikely to be driven by technological novelty alone; rather, it depends on a multidimensional mechanism involving usefulness, ease of use, credibility, and curricular embeddedness.

Existing medical education studies have begun to examine the use of LLMs in simulation-based training. Borg et al integrated an LLM with a social robotic platform to create more authentic virtual patient encounters for clinical reasoning training[39].

Öncü et al reported that a ChatGPT-4o–driven virtual standardized patient could be used to assess intern physicians’ competencies in clinical case management[40].

Wang et al showed that ChatGPT was feasible as a standardized patient in simulated medical assessment[41]. A recent scoping review of LLM-based virtual patients in medical education identified several limitations in current systems, including reliance on scripted and relatively narrow scenarios, limited use of real electronic health record data, simplified patient personalities and emotional styles, restricted multimodal interaction, inadequate longitudinal tracking of learner progress, and

limited support for modular training, tiered learning, intelligent feedback, and automated assessment[15]. In nursing education, research on LLM-VPs remains more limited. For example, Kim et al reported that GPT-based virtual patients may help novice nurses improve health assessment and communication abilities[17]. Overall, LLM-VP research in both medicine and nursing remains at an early stage. Existing studies have primarily examined immediate effects on communication skills or clinical competence, whereas less attention has been given to learners' needs, sustained adoption mechanisms, and the curricular conditions under which these systems may generate educational value. The expectations and concerns identified in this study should therefore be understood in relation to both the current stage of technological development and the unresolved challenges of educational implementation.

Notably, participants' anticipated attitudes toward LLM-VPs were accompanied by a clear awareness of boundaries and a strong capacity for risk recognition, rather than unidirectional technological optimism. Students recognized that virtual patients cannot fully replace the embodied presence, emotional complexity, and relational tension of real patient interactions. This is especially true in the cultivation of humanistic care and deep empathy, where authentic clinical contact remains irreplaceable. Some of these limitations may be alleviated through integration with more advanced technological ecosystems, such as social robots, mixed reality (MR), and virtual reality (VR), to support higher levels of interaction[39, 42]. Students were also highly sensitive to the accuracy and timeliness of knowledge content. This is unsurprising, as any incorrect professional information, fabricated content, or inappropriate communication guidance in nursing education may directly compromise

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learning quality and even distort clinical understanding[43, 44]. Future LLM-VP systems could incorporate retrieval-augmented generation (RAG)[45] and knowledge graphs (KGs)[46] to improve the accuracy, reasoning quality, and contextual relevance of system outputs. If real clinical cases are used to support system development, collaboration between developers and hospitals will be required to obtain authorized access to such data. Concerns related to privacy leakage may be partly addressed through local model deployment[47] or privacy-preserving large language model approaches based on adaptive compression[48]. Risks such as student overreliance on the system or gradual assimilation to AI-mediated communication patterns will still require active teacher guidance and broader institutional education. The central question is therefore not whether LLM-VPs can be introduced into nursing education, but whether they can be appropriately positioned, designed, validated, and governed.

Although this study was based on anticipated perceptions rather than actual user experience, it provides several preliminary implications for nursing education. First, LLM-VPs may be most appropriately positioned as supplementary and transitional training tools to support preclinical preparation, extracurricular reinforcement, and formative feedback, rather than as replacements for teacher guidance, standardized patients, or real patient contact. Second, the realization of their educational value depends on curricular integration rather than mere technical availability. To enhance sustained use and learning outcomes, LLM-VPs should be aligned with nurse-patient communication courses, objective structured clinical examination (OSCE) training, pre-internship preparation, and reflective assignments. Third, teachers remain central to this process and should be responsible for setting learning goals, interpreting

feedback, correcting errors, and guiding reflection so as to avoid students' mechanical dependence on system outputs. Finally, developers and administrators should recognize that students expect an educational product designed with nursing expert involvement, multiple levels of nursing-specific scenarios, actionable feedback, and mechanisms for error correction. Governance structures addressing content quality control, privacy protection, ethical review, and operational stability should be developed in parallel. Without such safeguards, the introduction of LLM-VPs may remain a conceptual demonstration rather than a substantive improvement in nursing education.

Limitations

This study has several limitations. First, participants were undergraduate nursing students recruited from a university in Hangzhou, China. The findings are therefore context dependent, and their transferability should be examined in other regions, educational systems, and cultural settings. Second, participants had not engaged in long-term or systematic use of a specific LLM-VP system. Their views were based primarily on prior understanding, expectations, and preliminary judgments about LLM-VPs. Accordingly, this study is better suited to identifying learner needs, adoption conditions, and implementation concerns than to evaluating actual educational effectiveness or sustained use behavior. Third, this study included only student perspectives and did not incorporate views from other stakeholders, such as nursing educators, clinical preceptors, administrators, or system developers. Finally, because LLM-related technologies are evolving rapidly, students' current perceptions may be time sensitive and should be reassessed as these systems mature and become more widely implemented.

Conclusions

This study showed that undergraduate nursing students held cautiously positive and clearly bounded expectations regarding LLM-based virtual patients. They did not view LLM-VPs as a replacement for real patient contact. Rather, they perceived them as a potential supplementary bridge between classroom learning and clinical practice that may help address limitations in current communication training, including insufficient authenticity, limited practice opportunities, delayed or inadequate feedback, and weak transfer to clinical settings. Students' anticipated willingness to use LLM-VPs was shaped by perceived usefulness, nursing relevance, curricular embeddedness, and system credibility. Future research should examine the feasibility, acceptability, and learning effects of LLM-VPs in authentic nursing education settings and should address implementation conditions such as knowledge verification, ethical governance, and teacher guidance to support cautious, standardized, and evidence-informed educational integration.

Declarations

List of abbreviations

AI: Artificial intelligence

LLMs: Large language models

VP: Virtual patient

EHRs: Electronic health records

COREQ: Consolidated Criteria for Reporting Qualitative Research

UTAUT: Unified Theory of Acceptance and Use of Technology

OSCE: Objective structured clinical examination

RAG: Retrieval-augmented generation

KGs: Knowledge graphs

Ethics approval and consent to participate

All methods were conducted in accordance with the ethical principles outlined in the Declaration of Helsinki. This study was approved by the Ethics Committee of School of Public Health and Nursing, Hangzhou Normal university (No. 2026009). Written informed consent was received from each participant prior to commencing interviews. The interviews were conducted in private rooms and, therefore, privacy was ensured. To ensure confidentiality, data were stored and handled securely to protect them from unauthorised access. For anonymity purposes, code names were assigned, and participant quotes are presented using code names.

Consent for publication

Not applicable.

Availability of data and materials

The anonymized transcripts supporting the findings of this study are available from the corresponding author upon reasonable request, in accordance with ethical approval and participant confidentiality agreements.

Competing interests

The authors declare that they have no competing interests

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Authors' contributions

JZ conceptualized the study, set the research methodology, and edited the manuscript. XZ conceptualized the study and revised the manuscript. XL, SL and SS organized the data and set the research methodology. QZ conceptualized the study, reviewed and edited the manuscript, and performed formal analysis. SC conceptualized the study, reviewed and edited the manuscript, acquired funding, managed the project, and performed formal analysis. All authors read and approved the final manuscript.

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Supplementary data

Supplementary Material 1. COREQ checklist.

Supplementary Material 2. Participant Interview Guide.

Supplementary Material 3. The participants illustrative quotations which are presented themes, subthemes and their illustrative quotes from participants.

References

1. Ha JF, Longnecker N: **Doctor-patient communication: a review.** *Ochsner J* 2010, **10**(1):38-43.
2. Tavakoly Sany SB, Behzad F, Ferns G, Peyman N: **Communication skills training for physicians improves health literacy and medical outcomes among patients with hypertension: a randomized controlled trial.** *BMC Health Serv Res* 2020, **20**(1):60.
3. Kolb DA: **Experience as the source of learning and development.** *Upper Sadle River: Prentice Hall* 1984:1554-1573.

4. Kolb DA: **Experiential learning: Experience as the source of learning and development**: FT press; 2014.
5. He H, Xu XH, Gu YY, Duan YP, Li SM, Vesga JB: **Preparing for meeting patients: A generative AI-enhanced virtual reality patient encounter practice**. *Clinical Simulation in Nursing* 2026, **111**.
6. Jo YW, Lee M, Yang HJ: **Large Language Model-Based Virtual Patient Simulations in Medical and Nursing Education: A Review**. *Applied Sciences-Basel* 2025, **15**(22).
7. Choi H, Lee U, Gwon T: **Development of a Computer Simulation-based, Interactive, Communication Education Program for Nursing Students**. *Clinical Simulation in Nursing* 2021, **56**:1-9.
8. Holderried F, Stegemann-Philipps C, Herschbach L, Moldt JA, Nevins A, Griewatz J, Holderried M, Herrmann-Werner A, Festl-Wietek T, Mahling M: **A Generative Pretrained Transformer (GPT)-Powered Chatbot as a Simulated Patient to Practice History Taking: Prospective, Mixed Methods Study**. *Jmir Medical Education* 2024, **10**.
9. Vessey JA, Huss K: **Using standardized patients in advanced practice nursing education**. *Journal of Professional Nursing* 2002, **18**(1):29-35.
10. Gelis A, Cervello S, Rey R, Llorca G, Lambert P, Franck N, Dupeyron A, Delpont M, Rolland B: **Peer Role-Play for Training Communication Skills in Medical Students A Systematic Review**. *Simulation in Healthcare-Journal of the Society for Simulation in Healthcare* 2020, **15**(2):106-111.
11. Hickman C, Jennings M, Smith TS: **Simulating Patient Encounters With Generative Artificial Intelligence Effective Prompt Engineering Techniques**. *Nurse Educator* 2025, **50**(1):E53-E54.
12. Naamati-Schneider L, Alt D: **Developing AI literacy in medical and health professions education: Exploring ChatGPT, ChatBots development, and simulation-based learning**. *Nurse Education in Practice* 2025, **88**.
13. Stanceski K, Zhong SR, Zhang XM, Khadra S, Tracy M, Korla L, Lo SRT, Naganathan V, Kim J, Dunn AG *et al*: **The quality and safety of using generative AI to produce patient-centred discharge instructions**. *Npj Digital Medicine* 2024, **7**(1).
14. Xu HL, Chen PH, Luo ZY, Chen YE, He PJ, Tan YB: **"Is GenAI my supervisor?" Experiences of nursing undergraduates in using generative AI to assist in case study theses: A qualitative study**. *Nurse Education Today* 2026, **162**.
15. Zeng J, Qi W, Shen S, Liu X, Li S, Wang B, Dong C, Zhu X, Shi Y, Lou X *et al*: **Embracing the Future of Medical Education With Large Language Model-Based Virtual Patients: Scoping Review**. *Journal of Medical Internet Research* 2025, **27**.
16. Benfatah M, Marfak A, Saad E, Hilali A, Nejari C, Youlyouz-Marfak I: **Assessing the efficacy of ChatGPT as a virtual patient in nursing simulation training: A study on nursing students' experience**. *Teaching and Learning in Nursing* 2024, **19**(3):e486-e493.

17. Kim J, Won J, Lee Y: **Use of a generative pre-trained transformer-based virtual patient for health assessment and communication training in nursing education: A mixed-methods study.** *Nurse Education in Practice* 2025, **88**.
18. Sandelowski M: **Whatever happened to qualitative description?** *Research in nursing & health* 2000, **23**(4):334-340.
19. Adams A, Cox AL: **Questionnaires, in-depth interviews and focus groups:** Cambridge University Press; 2008.
20. Tong A, Sainsbury P, Craig J: **Consolidated criteria for reporting qualitative research (COREQ): a 32-item checklist for interviews and focus groups.** *International Journal for Quality in Health Care* 2007, **19**(6):349-357.
21. Baltacı A: **Nitel araştırma süreci: Nitel bir araştırma nasıl yapılır?** *Ahi Evran Üniversitesi Sosyal Bilimler Enstitüsü Dergisi* 2019, **5**(2):368-388.
22. Krueger RA, Casey MA: **Focus Group Interviewing.** In: *Handbook of Practical Program Evaluation: Fourth Edition.* edn.: Wiley Blackwell; 2015: 506-534.
23. Braun V, Clarke V: **Thematic analysis: A practical guide.** 2021.
24. Goundar PR: **Researcher Positionality: Ways to Include it in a Qualitative Research Design.** *Int J Qual Method* 2025, **24**.
25. Lincoln YS: **Naturalistic inquiry,** vol. 75: sage, 1985.
26. Olmos-Vega FM, Stalmeijer RE, Varpio L, Kahike R: **A practical guide to reflexivity in qualitative research: AMEE Guide No. 149.** *Medical teacher* 2023, **45**(3):241-251.
27. Daneshfar M, Moonaghi HK: **The impact of clinical simulation on bridging the theory-practice gap in nursing education: a systematic review.** *Bmc Medical Education* 2025, **25**(1).
28. Gaba DM: **The future vision of simulation in health care.** *Quality & safety in health care* 2004, **13 Suppl 1**:i2-10.
29. Liao G, Liu Z, Zhang Z, Shang J, Yi Y, Xu D, Wan H, Yao J: **Differences Between Preceptors and Nursing Students in Understanding Teaching Needs <i>A Mixed Methods Study</i>.** *Nurse Educator* 2026.
30. Oestensson S, Mora MA, Darcy L, Huttu LS, Van Dulmen S, Sundler AJ: **Patient-Registered Nurse Communication: Exploring Patients' Expectations and Experience in Primary Care Consultations.** *Health Expectations* 2025, **28**(4).
31. Grant JB, Jones J, Candrian C, Oman KS, Reed SM: **The role of nursing communication: A critical interpretive synthesis.** *International Journal of Nursing Studies Advances* 2025, **9**.
32. Stratton-Maher D, Tulleners T, Elliott J, Ryan L, McCulloch J, Roderick G, Jayasinghe T, Buckley J, Newman J-M, Nutter H et al: **Re-imagining nursing education: Mapping student learning, transition, and workforce readiness-A scoping review.** *International Journal of Nursing Studies* 2026, **178**.
33. Venkatesh V, Morris MG, Davis GB, Davis FD: **User acceptance of information technology: toward a unified view.** *MIS Quarterly* 2003, **27**(3):425-478.

34. Dogham RS, Ali HFM, Ghaly AS, Elcokany NM, Seweid MM, El-Ashry AM: **Deciphering the influence: academic stress and its role in shaping learning approaches among nursing students: a cross-sectional study.** *Bmc Nursing* 2024, **23**(1).
35. Mohamed NA, Ali SO, Ehrahim EEE, Ahmed AL, Wahba AM: **Predictors of Academic and Clinical Stress Among Nursing Students.** *Sage Open Nursing* 2024, **10**.
36. Kang K, Yu M: **Rapid cycle deliberate practice simulation with standardized prebriefing and video based formative feedback in advanced cardiac life support.** *Scientific Reports* 2025, **15**(1).
37. Jawabreh N, Hamdan-Mansour A, Harazne L, Ayed A: **Effectiveness of high-fidelity simulation on practice, satisfaction, and self-confidence among nursing students in mental health nursing class.** *Bmc Nursing* 2025, **24**(1).
38. Toqan D, Ayed A, Khalaf IA, Alsadi M: **Effect of High-Fidelity Simulation on Self-Satisfaction and Self-Confidence Among Nursing Students.** *Sage Open Nursing* 2023, **9**.
39. Borg A, Georg C, Jobs B, Huss V, Waldenlind K, Ruiz M, Edelbring S, Skantze G, Parodis I: **Virtual Patient Simulations Using Social Robotics Combined With Large Language Models for Clinical Reasoning Training in Medical Education: Mixed Methods Study.** *Journal of Medical Internet Research* 2025, **27**.
40. Oncu S, Torun F, Ulku HH: **AI-powered standardised patients: evaluating ChatGPT-4o's impact on clinical case management in intern physicians.** *Bmc Medical Education* 2025, **25**(1).
41. Wang C, Li S, Lin N, Zhang X, Han Y, Wang X, Liu D, Tan X, Pu D, Li K et al: **Application of Large Language Models in Medical Training Evaluation-Using ChatGPT as a Standardized Patient: Multimetric Assessment.** *Journal of Medical Internet Research* 2025, **27**.
42. Maquilón RG, Uhl J, Schrom-Feiertag H, Tscheligi M: **Integrating GPT-Based AI into Virtual Patients to Facilitate Communication Training Among Medical First Responders: Usability Study of Mixed Reality Simulation.** *Jmir Formative Research* 2024, **8**.
43. Topaz M, Peltonen LM, Michalowski M, Stiglic G, Ronquillo C, Pruinelli L, Song JY, O'Connor S, Miyagawa S, Fukahori H: **The ChatGPT Effect: Nursing Education and Generative Artificial Intelligence.** *Journal of Nursing Education* 2025, **64**(6):e40-e43.
44. Iqbal U, Tanweer A, Rahmanti AR, Greenfield D, Lee LTJ, Li YCJ: **Impact of large language model (ChatGPT) in healthcare: an umbrella review and evidence synthesis.** *Journal of Biomedical Science* 2025, **32**(1).
45. Miao Y, Zhao Y, Luo Y, Wang H, Wu Y: **Improving Large Language Model Applications in the Medical and Nursing Domains With Retrieval-Augmented Generation: Scoping Review.** *Journal of Medical Internet Research* 2025, **27**.
46. Murali L, Gopakumar G, Viswanathan DM, Raman R, Nedungadi P: **Integrating LLMs and Knowledge Graphs for Medical AI: Advances, Challenges, and Future Directions.** *IEEE journal of biomedical and health informatics* 2025, **PP**.

47. Montagna S, Ferretti S, Klopfenstein LC, Florio A, Pengo MF: **Data Decentralisation of LLM-Based Chatbot Systems in Chronic Disease Self-Management**; 2023.
48. Gong X, Gao J, Sun S, Zhong Z, Shi Y, Zeng H, Yang K: **Adaptive Compressed-based Privacy-preserving Large Language Model for Sensitive Healthcare**. *IEEE journal of biomedical and health informatics* 2025, **PP**.

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